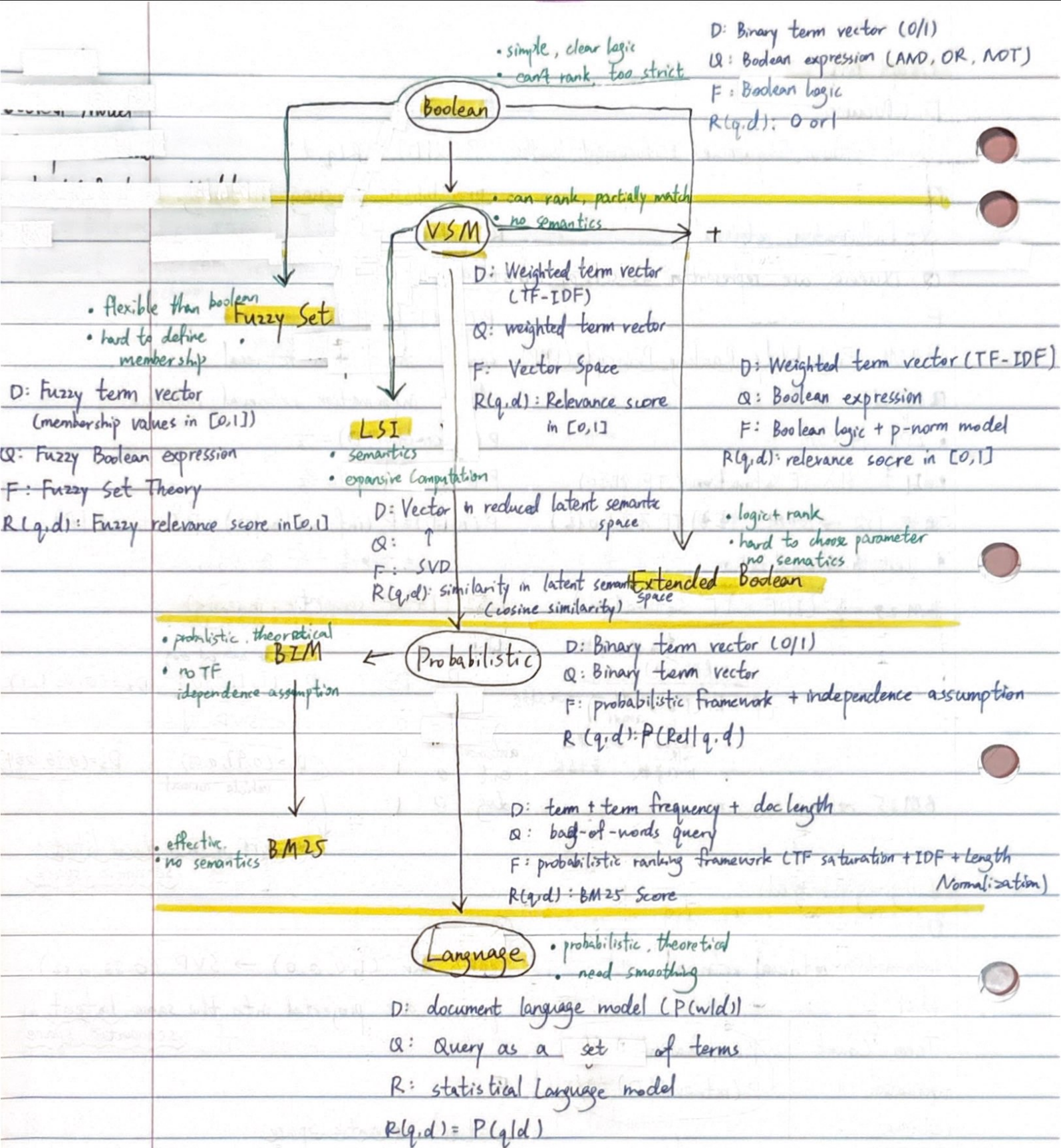




Relevant / Not relevant

• Precision / Recall

↓
 返回5篇相关3篇
 $P = \frac{3}{5} = 0.6$

↓
 有10篇相关找到5篇
 $R = \frac{5}{10} = 0.5$

• F-measure

$$F_1 = \frac{2PR}{P+R} = \frac{2 \times 0.6 \times 0.5}{0.6 + 0.5} = \frac{0.6}{1.1} = 0.55$$

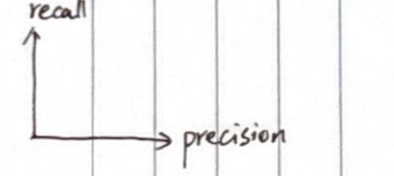
Ranking Evaluation

• Precision @ N

top 10, 7篇相关
 $P@10 = \frac{7}{10}$

RRRNNNNNNN
 (NNNRRRRRRR) 改进

• Precision-Recall Curve



• R-Precision

有R个相关文档
 只看前R个结果

• Average Precision (AP)

$Q_1: RNRNR$
 每找到一个相关文档, 记录P
 $P@1 = 1, P@3 = \frac{2}{3}, P@5 = \frac{3}{5}$
 $(P@1 + P@3 + P@5) / 3$

• Mean Average Precision (MAP)

多个查询取平均
 $Q_1, Q_2, Q_3 \quad MAP = \text{mean}(AP)$

• User-Oriented Ranking

用户只看第一

• Mean Reciprocal Rank (MRR)

$Q_1: RNN \quad RR_1 = \frac{1}{1}$ 第一个相关文档出现在 rank 1
 $Q_2: NRN \quad RR_2 = \frac{1}{2}$
 $Q_3: NNNR \quad RR_3 = \frac{1}{4}$
 $MRR = \text{Mean}(RR) = \frac{(RR_1 + RR_2 + RR_3)}{3}$

• Discounted Cumulative Gain (DCG)

DCG 排名越前越好, 越后 discount
 $rank \quad relevance \quad gain$
 $1 \quad 3 \quad DCG = \sum \frac{gain}{\log_2(rank+1)}$
 $rank 1: \frac{3}{\log_2(2)} = 3$
 $rank 2: \frac{2}{\log_2(3)} = 1.262$
 $rank 3: \frac{1}{\log_2(4)} = 0.5$
 $DCG = 3 + 1.26 + 0.5 = 4.762$

• Ideal DCG

把文档按 gain 从高到低排
 $ND CG = \frac{DCG}{IDCG}$

0 ~ 1
 坏 完美

• Expected Reciprocal Rank (ERR)

$MRR + \text{User Satisfaction} \rightarrow ERR$
 用户平均在 rank 几找到满意答案

$R(g) = \frac{2^g - 1}{2^g \max}$

relevance / gain	satisfaction	Probability
0	0	0.125
1	0.375	0.375
2	0.875	0.875

$$ERR = \sum_{r=1}^n \frac{1}{r} P \quad (\text{用户第一次在 } r \text{ 停止})$$

rank	relevance	satisfaction P
1	2	0.375
2	3	0.875

rank 1 就满足: $P_1 = 0.375 \quad ERR_1 = \frac{1}{1} \times 0.375$

rank 2 满足: 用户看到 rank 2: $1 - 0.375 = 0.625$

$0.625 \times P_2 = 0.625 \times 0.875 = 0.547$

$$ERR_2 = \frac{1}{1+1} \times 0.547$$

• Diversity

• Maximal Marginal Relevance (MMR)

既要相关, 又不要和已经选中的结果太像

relevance + diversity

	relevance	similarity
D_2	0.8	0.7
D_3	0.7	0.1

$$D_2: 0.8 \times 0.5 - 0.7 \times 0.5 = 0.05$$

$$D_3: 0.7 \times 0.5 - 0.1 \times 0.5 = 0.30$$

Pearson $r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2} \sqrt{\sum (y_i - \bar{y})^2}}$
 $r = 0.7$ 评分变化趋势相近

Ranking Comparison

• Spearman Rank Correlation

Doc	Rank A	Rank B	d_i	d_i^2
D_1	1	1	0	0
D_2	2	3	-1	1
D_3	3	2	1	1
D_4	4	4	0	0

$d_i = \text{rank A} - \text{rank B}$

$$\sum d_i^2 = 2$$

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} = 1 - \frac{6 \times 2}{4(16 - 1)} = 0.8$$

$-1 \leq \rho \leq 1$ 一样
 相反
 0 没关系

• Kendall Tau

A: $D_1 D_2 D_3$
 B: $D_1 D_3 D_2$

比较 (D_1, D_2)

(D_1, D_3) 一致 C

(D_2, D_3) 不一致 D

$$\tau = \frac{C - D}{\frac{n(n-1)}{2}}$$

$-1 \leq \tau \leq 1$
 同上

• Pearson product-moment correlation

coefficient	Alice	Bob	Alice 评分	Bob 评分
movie A	5	4	1.75	0.75
movie B	3	2	-0.25	-1.25
movie C	4	5	0.75	1.75
movie D	1	2	-2.25	-1.25

$\bar{x} = 3.25 \quad \bar{y} = 3.25$

Relevance Feedback

explicit relevance feedback

① user
↳ search: information retrieval

② system return

↳ doc content
D1 information retrieval
D2 search engine
D3 database
D4 operating system

③ user:

↳ doc judgement
D1 R
D2 R
D3 N
D4 N

④ system

↳ initial query: information retrieval
R doc: search, ranking, index, query

Query expansion:

- accurate (fold by users themselves)
high-quality labels
professional search
- ~~big~~ nobody wants to label
subjective (different opinions)
feedback isn't enough

implicit relevance feedback

Click 点击

Dwell Time 停留时间

- user does nothing
large-scale
real user behavior
- noisy (click ≠ relevant)
less accurate
position bias
(users are more likely to click the first position)

Download

Bookmark/Favorite

Purchase

ATA

Pseudo Relevance Feedback

① user

↳ search: information retrieval

② system return (BM25)

↳ rank doc
1 D1
2 D2
3 D3
4 D4
5 D5

③ system → suppose Top3 Relevant

④ system count high-frequency words

word	freq
search	20
ranking	18
query	15
index	12

⑤ system

Query expansion:

- information retrieval
- + search, ranking, query, index
- user does nothing
- query drift
(wrong feedback
→ wrong direction)
dependent on initial results
error amplification

Query Expansion

① synonym expansion 同义词扩展
car → car, automobile, vehicle

② wordNet 词典扩展
wordNet knows that "dog animal
canine, puppy" have relationship
↳ expand automatically

③ relevance feedback ^{based} expansion

④ pseudo relevance feedback

⑤ LSI

↳ recall ↑
precision ↓ (noisy
query drift)

relevance feedback of

Rocchio Algorithm (VSM)

search: information retrieval

$$Q = (1, 1)$$

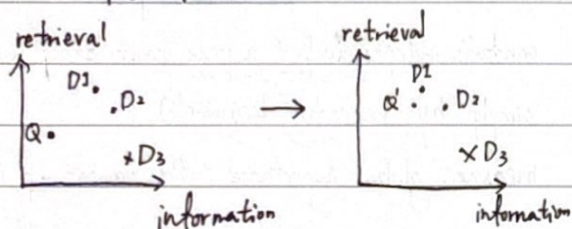
suppose term space only has:

t_1 = information, t_2 = retrieval
system return

doc

D1 R
D2 R
D3 N

let Q get near to R
far from N



1. Query: $Q = (1, 1)$

Relevant: $D_1 = (2, 3)$

$D_2 = (3, 2)$

Non-Relevant: $D_3 = (5, 0)$

参数: $\alpha = 1$

$\beta = 0.75$

$\gamma = 0.25$

$$\alpha Q = (1, 1)$$

$$\beta \frac{1}{|D_r|} \sum D_r = 0.75 \cdot \left(\frac{(2, 3) + (3, 2)}{2} \right) = (1.875, 1.875)$$

$$\gamma \frac{1}{|D_n|} \sum D_n = 0.25 (5, 0) = (1.25, 0)$$

$$Q_{\text{new}} = (1, 1) + (1.875, 1.875) - (1.25, 0) = (1.625, 2.875)$$

• simple, interpretable (can see R improve and N decrease)

• need feedback, query drift, single topic assumption
improve
use top-ranked irrelevant document instead of all irrelevant documents

$$Q_{\text{new}} = \alpha Q + \beta \frac{1}{|D_r|} \sum_{d \in D_r} d - \gamma \frac{1}{|D_n|} \sum_{d \in D_n} d$$

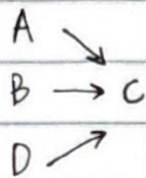
αQ : user's initial query

$\beta \frac{1}{|D_r|} \sum D_r$: relevant centroid (相关文档中心)
 $\frac{D_1 + D_2}{2}$

$\gamma \frac{1}{|D_n|} \sum D_n$: non-relevant Documents centroid

PageRank

independent on topic (只计算那些权重, 所有 topic 通用)
global importance



$$PR(A) = \sum_{i \in In(A)} \frac{PR(i)}{L(i)}$$

$PR(A)$: 页面 A 的 PageRank

$In(A)$: 所有指向 A 的页面集合

$PR(i)$: 页面 i 的 PageRank

$L(i)$: 页面 i 的出链数 (Out-degree)

second round

$$PR(A) = 0.05 + \frac{PR(C)}{L(C)} = 1.6$$

$$PR(B) = 0.05 + \frac{PR(A)}{L(A)} = 0.575$$

$$PR(C) = 0.05 + \frac{PR(A)}{L(A)} + \frac{PR(B)}{L(B)} = 0.05 + 0.525 + 0.55 = 1.125$$

Node PR

A	1.6
B	0.575
C	1.125

引入 Damping Factor?

$A \rightarrow B$ OutDegree(B) = 0

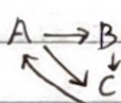
随机冲浪者 85% 按链接点击, 15% 随机到网页
random surfer

$$PR(A) = \frac{1-d}{N} + d \sum_{i \in In(A)} \frac{PR(i)}{L(i)} \quad d = 0.85$$

↓ ↓
随机跳转 链接投票

Example:

$A \rightarrow B$
 $A \rightarrow C$
 $B \rightarrow C$
 $C \rightarrow A$



step 1: $L(A) = 2, L(B) = 1, L(C) = 1$

step 2: Initial PR $PR(A) = 1, PR(B) = 1, PR(C) = 1$

step 3:

$$PR(A) = \frac{1-d}{N} + \sum \frac{PR(i)}{L(i)}$$

$$= \frac{1-0.85}{3} + \frac{PR(C)}{L(C)} = 0.05 + 1 = 1.05$$

$$PR(B) = \frac{1-d}{N} + \sum \frac{PR(i)}{L(i)}$$

$$= 0.05 + \frac{PR(A)}{L(A)} = 0.05 + 0.5 = 0.55$$

$$PR(C) = \frac{1-d}{N} + \sum \frac{PR(i)}{L(i)}$$

$$= 0.05 + \frac{PR(A)}{L(A)} + \frac{PR(B)}{L(B)} = 1.55$$

→ first round

Node	PR
A	1.05
B	0.55
C	1.55

- uses link structure (wiki important, many pages link to it)
- content-independent (a page can't easily improve its PageRank simply by repeating keywords)
- measure global importance (the number of incoming links + the importance of the linking pages)

① Ignores Query Topic (search: ML → google homepage)

Bias Against New Pages

② vulnerable to Link Spam (pages link to each other)
expensive computation (iterative until convergence)

↓ improvements

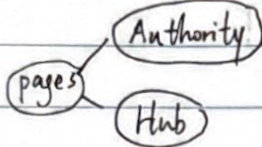
① topic-sensitive pageRank (random Jump → topic Pages only)

② personalized PageRank (global importance → personal importance)

③ trustRank

reduces link spam by propagating trust from a set of manually selected trustworthy pages)

HITS



每个网页都有 Authority score : a_i

Hub score : h_i

Authority Update: $a_i = \sum_{j \rightarrow i} h_j$

Hub Update: $h_i = \sum_{i \rightarrow j} a_j$

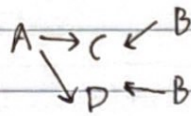
Example:

$A \rightarrow C$

$A \rightarrow D$

$B \rightarrow C$

$B \rightarrow D$



step 1: initialize $a_i=1, h_i=1$

step 2: authority

$$a_c = h_a + h_b = 1 + 1 = 2$$

$$a_d = h_a + h_b = 1 + 1 = 2$$

$$a_a = 0$$

$$a_b = 0$$

step 2: hub

$$h_a = a_c + a_d = 2 + 2 = 4$$

$$h_b = a_c + a_d = 2 + 2 = 4$$

$$h_c = 0$$

$$h_d = 0$$

page	authority	hub
A	0	4
B	0	4
C	2	0
D	2	0

step 3: normalization

$$a'_i = \frac{a_i}{\sum a_i} \quad h'_i = \frac{h_i}{\sum h_i}$$

$$a_i = \frac{a_i}{\sqrt{8}} \quad h_i = \frac{h_i}{\sqrt{32}}$$

Page	authority	hub
A	0	0.707
B	0	0.707
C	0.707	0
D	0.707	0

• Hub-Authority Distinction

top-k

topic-oriented (query $\rightarrow$ root set $\rightarrow$ hits)

interpretable (good hubs or authorities)

① topic drift (root set contains irrelevant pages)

② vulnerable to link spam

expensive at Query Time (HITS must be computed for each query)

③ sensitive to the root set

improvements

require expansion

① improve root set (root set)

② weighted HITS (trusted page $\rightarrow$ high weight $\rightarrow$ low weight)

③ Anti link spam

④ content-aware HITS

(query $\rightarrow$ root set $\rightarrow$ hits)

↑
bm25

$$a_i = a_i + \lambda \times \text{sim}(\text{Query}, \text{Page})$$

$$A \rightarrow B \quad A \times B \quad \text{weight}(A, B) = 0.9$$

$$a(B) = \sum \text{weight}(i, B) \times \text{Hub}(i)$$

Complex Network

Scale-free network

degree distribution

k	节点数	→	k	P(k)
1	500		1	0.55
2	300		2	0.33
3	100		3	0.11
100	1		100	0.001

$$\text{Var}(k) = E(k^2) - E(k)^2$$

$$E(k^2) = \int k^2 (ck^{-r}) dk$$

$$= \int ck^{2-r} dk$$

$$2-r < -1 \quad r > 3$$

$$P(k) = c$$

$$P(k) = ck^{-r}$$

scale-free network

Power Law (幂律分布)

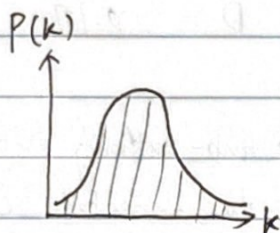
例: $P(k) = k^{-2}$

$k=1 \quad P(1)=1$

$k=2 \quad P(2)=\frac{1}{4}$

$k=10 \quad P(10)=\frac{1}{100}$

$k=100 \quad P(100)=\frac{1}{10000}$



r	E(k)	Var(k)
$r > 3$	收敛	收敛
$2 < r \leq 3$	收敛	发散
$r \leq 2$	发散	发散

$$\int P(k) dk = 1 \rightarrow \int_{k_{min}}^{\infty} ck^{-r} dk = 1$$

例: $P(k) = ck^{-3} \quad k \geq 1$

$$\int_1^{\infty} ck^{-3} dk = 1$$

$$\left[-\frac{1}{2} k^{-2} \cdot c \right]_1^{\infty} = 1$$

$$\lim_{k \rightarrow \infty} \left(-\frac{1}{2} k^{-2} \cdot c \right) - \left(-\frac{1}{2} (1)^{-2} \cdot c \right) = 1$$

$$\frac{1}{2} c = 1, \quad c = 2$$

$$P(k) = 2k^{-3}$$

$$E(k) = \int k P(k) dk \rightarrow E(k) = \int k (ck^{-r}) dk = c \int k^{1-r} dk$$

例: $E(k) = \int_1^{\infty} k \cdot 2k^{-3} dk$

$$= 2 \int_1^{\infty} k^{-2} dk = 2 \left[-k^{-1} \right]_1^{\infty} = 2$$

收敛: $1-r < -1, \quad r > 2$

Small World = short average path length + high clustering

distance = shortest path length

A-B-C-D-E

$$d(A, E) = 4$$

Cluster = tightly connected local group of nodes

community = A group of nodes with dense internal connections and sparse external connections

Closeness Centrality 离所有节点有多近

A-B-C-D-E

$$cc(v) = 1 / \sum d(v, u)$$

$$cc(C) = \frac{1}{d(C, A) + d(C, B) + d(C, D) + d(C, E)}$$

$$= \frac{1}{2 + 1 + 2 + 1} = \frac{1}{6}$$

$$cc(A) = 1 / \sum d(v, u)$$

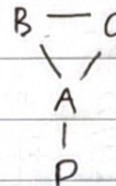
$$cc(A) = \frac{1}{d(A, B) + d(A, C) + d(A, D) + d(A, E)}$$

$$= \frac{1}{1 + 2 + 3 + 4} = \frac{1}{10}$$

距大家越近, cc 越大

Cluster Coefficient 邻居间有多紧密

$$C_i = \frac{E_i}{\frac{k_i(k_i-1)}{2}} \quad \begin{array}{l} k_i: \text{节点 } i \text{ 的 degree} \\ E_i: \text{邻居之间实际存在的边数} \end{array}$$



$$C_A = \frac{E_A}{\frac{k_A(k_A-1)}{2}} = \frac{1}{\frac{3(3-1)}{2}} = \frac{1}{3}$$

Betweenness Centrality 有多少最短路径经过该

A-B-C-D-E

$$BC(v) = \sum \frac{\sigma_{st}(v)}{\sigma_{st}} \quad \begin{array}{l} \downarrow \quad \downarrow \\ \text{number of shortest paths (through } v) \quad \text{number of shortest paths} \end{array}$$

$$BC(C) = \frac{\sigma_{AB}(C)}{\sigma_{AB}} + \frac{\sigma_{AD}(C)}{\sigma_{AD}} + \frac{\sigma_{AE}(C)}{\sigma_{AE}} + \frac{\sigma_{BD}(C)}{\sigma_{BD}} + \frac{\sigma_{BE}(C)}{\sigma_{BE}} + \frac{\sigma_{DE}(C)}{\sigma_{DE}}$$

A B D E

(A, B) (A, D) (A, E) (B, D) (B, E) (D, E)

$$(A, B) \frac{\sigma_{AB}(C)}{\sigma_{AB}} = 0$$

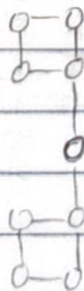
$$(A, D) \frac{\sigma_{AD}(C)}{\sigma_{AD}} = \frac{1}{1} = 1$$

$$(A, E) \frac{\sigma_{AE}(C)}{\sigma_{AE}} = \frac{1}{1} = 1$$

$$(B, D) \frac{\sigma_{BD}(C)}{\sigma_{BD}} = \frac{1}{1} = 1$$

$$(B, E) \frac{\sigma_{BE}(C)}{\sigma_{BE}} = \frac{1}{1} = 1$$

$$(D, E) \frac{\sigma_{DE}(C)}{\sigma_{DE}} = 0$$



BC 大

连接两个团体

$$BC(C) = 0 + 1 + 1 + 1 + 1 + 0 = 4$$

Connected Graph

任意两点间存在距离

A-B-C-D

dis connected graph

A-B

C-D

Complete Graph

K_n , 任意两个点之间都有边

K_4 A-B
1 X 1
C-D

• degree

K_n : $\deg(v) = n-1$, K_4 : $\deg = 3$

• distance

都是 $d(u, v) = 1$

• CC

都相同 $CC(A) = \frac{1}{d(A, C) + d(A, B) + d(A, D)}$
 $= \frac{1}{1+1+1} = \frac{1}{3}$

• BC

所有点之间都可直接到达, 不依赖中间节点, $BC = 0$

• C_i

$C_i(A) = \frac{3}{3(3-1)} = 1$ 都是 1

Star Graph

B
|
C-A-D
|
E

• degree 中心最大

中心: $\deg(A) = 4$

叶子: $\deg = 1$

• distance

中心: $\deg = 1$

叶子: $\deg = 2$

• CC 中心最大

中心: $CC(A) = \frac{1}{d(A, B) + d(A, C) + d(A, D) + d(A, E)}$
 $= \frac{1}{1+1+1+1} = \frac{1}{4}$

叶子: $CC(B) = \frac{1}{d(B, A) + d(B, C) + d(B, D) + d(B, E)}$
 $= \frac{1}{1+2+2+2} = \frac{1}{7}$

• BC 中心最大

中心: $BC(A) = 0$

叶子: $BC(B) = 0$

• C_i

中心: $C_i(A) = 0$ 都是 0

叶子: $C_i(B) = 0$

Ring Graph

A-B
| |
E C
| |
D

• degree 都一样

$\deg = 2$

• distance

$d(A, B) = 1$

$d(A, C) = 2$

$d(A, D) = 2$

• CC 都一样

$CC = \frac{1}{1+2+2+1} = \frac{1}{6}$

• BC 都一样

$BC(A)$

$(B, C) (B, D) (B, E) (C, E) (D, E)$

$\frac{\sigma_{BC}(A)}{\sigma_{BC}} = 0$

$\frac{\sigma_{BD}(A)}{\sigma_{BD}} = 0$

$\frac{\sigma_{BE}(A)}{\sigma_{BE}} = 1$

$\frac{\sigma_{CE}(A)}{\sigma_{CE}} = 0$

$\frac{\sigma_{DE}(A)}{\sigma_{DE}} = 0$

$BC(A) = 1$

• $C_i \geq 0$

$C_i = 0$

Chain Graph

A-B-C-D-E

• degree

$\deg(A) = \deg(E) = 1$

中间 $\deg = 2$

• distance

$d(A, E) = 4$

• CC 中间最大

中心: $CC(C) = \frac{1}{1+2+1+2} = \frac{1}{6}$

$CC(A) = \frac{1}{1+2+3+4} = \frac{1}{10}$

• BC 中间最大

$BC(C) = 4$

$BC(A) = 0$

$BC(B) = 3$

$BC(D) = 3$

$BC(E) = 0$

• $C_i \geq 0$

$C_i = 0$